Taylor and Francis Book Chapter

LOC Page

Contents

CHAPTER 1 ▪ Optimizing Feature Selection and Model Accuracy on Large Datasets: A Hybrid Approach with TabMap 1

SAKIB HOSSAIN, BADHON NATH JOY, TANJIB UL ISLAM, SUBRINA ISLAM PRITY , and TOHEDUL ISLAM*

1.1 INTRODUCTION 2

1.2 RELATED WORKS 4

1.3 METHODS 6

1.3.1 Research Methodology 6

1.3.2 Dataset and Data Preprocessing 6

1.3.3 Baseline Machine Learning Model Training 7

1.3.4 TabMap Spatial Transformation 7

1.3.5 TabMap Representation 8

1.3.6 Feature Selection 8

1.3.7 CNN Architecture 9

1.3.8 Model Evaluation 9

1.4 RESULTS OR FINDINGS 9

1.4.1 Performance Comparison: Traditional Models versus CNN on TabMap Data 10

1.4.2 Impact of Feature Selection on CNN Performance 10

1.5 DISCUSSION 11

1.6 CONCLUSION AND FUTURE WORK 12

1.7 ACKNOWLEDGEMENT 13

CHAPTER 1

Optimizing Feature Selection and Model Accuracy on Large Datasets: A Hybrid Approach with TabMap

Sakib Hossain

Department of Computer Science, American International University-Bangladesh, Dhaka 1229, Bangladesh, 21-45388-3@student.aiub.edu

Badhon Nath Joy

Department of Computer Science, American International University-Bangladesh, Dhaka 1229, Bangladesh, 21-45390-3@student.aiub.edu

Tanjib Ul Islam

Department of Computer Science, American International University-Bangladesh, Dhaka 1229, Bangladesh, 21-45393-3@student.aiub.edu

Subrina Islam Prity

Department of Computer Science, American International University-Bangladesh, Dhaka 1229, Bangladesh, 21-45813-3@student.aiub.edu

Tohedul Islam*

Department of Computer Science, American International University-Bangladesh, Dhaka 1229, Bangladesh, islamtohedul@aiub.edu

CONTENTS

1.1 Introduction 2
1.2 Related Works 4
1.3 Methods 6
1.3.1 Research Methodology 6
1.3.2 Dataset and Data Preprocessing 6
1.3.3 Baseline Machine Learning Model Training 7
1.3.4 TabMap Spatial Transformation 7
1.3.5 TabMap Representation 8

1.3.6 Feature Selection 8
1.3.7 CNN Architecture 9
1.3.8 Model Evaluation 9
1.4 Results or Findings 9
1.4.1 Performance Comparison: Traditional Models versus CNN on TabMap Data 10
1.4.2 Impact of Feature Selection on CNN Performance 10
1.5 Discussion 11
1.6 Conclusion and Future Work 12
1.7 Acknowledgement 13

A novel hybrid approach is proposed for enhancing the classification of large tabular datasets by combining feature selection and spatial transformation using the TabMap algorithm, in combination with convolutional neural networks (CNNs). Traditional machine learning (ML) models have the shortcoming of not being able to learn complex dependencies among features in high-dimensional tabular data. Tabular data was transformed into spatially ordered 2D images using TabMap, such that feature correlations were conserved and local patterns were utilisable by CNNs for enhancing classification. The technique was tested using a large freshwater quality dataset of 4.5 million instances and 23 physicochemical attributes. After using preprocessing steps of imputation and normalisation, performance of CNN models on TabMap images fared comparably to traditional ML models such as Random Forest and XGBoost. Feature selection using Information Gain-based thresholding and Correlation-Based Feature Selection (CBFS) was also adopted for enhancing model efficiency. Results showed that CNN models on TabMap-transfomed data significantly outperformed traditional models by noteworthy improvements in accuracy and F1 score. Feature selection was also shown to enhance model interpretability and enable dimension reduction. The synergy of feature selection and spatial transformations for enhancing the classification of large environmental data is noted, providing direction for future investigations on enhancing model extendibility and model generalizability.

1.1 INTRODUCTION

In today's world of big data, vast amounts of information are generated daily in a wide range of fields such as environmental monitoring, healthcare, financial markets, and scientific discoveries. A significant volume of such information is in tabular form, with samples indicated by rows and different features by columns. While the presentation seems simple, analysis of table-based data proves to be greatly challenging. The highly involved interdependencies and non-linear relationships among features often hide amidst the data, rendering traditional machine learning (ML) models incapable of harnessing the inherent information fully. As described by Arik and Pfister, approaches of deep learning have not fared well against classical ML approaches in the analysis of table-based information, due mostly to these complex feature interactions and the lack of good representation methodologies [1].

Despite the notable strides made on artificial intelligence and deep learning models, issues persist in yielding high accuracy levels in massive tabular dataset spaces, especially when such dataset spaces cross multiple application fields. The classical models have the limitation of not effectively learning intricate feature dependencies due to the lack of adequate representation methodologies. The present study aims to bridge the gap by converting tabular data into spatially interpretable two-dimensional image spaces by virtue of the TabMap algorithm, in conjunction with Convolutional Neural Networks (CNNs). The primary research question is to analyze the performance of spatial encoding methodologies in enhancing model performance for massive tabular data. The effectiveness of transforming tabular data into spatial representations to leverage CNNs has been highlighted in several studies, such as in TabMap [9] and DeepInsight [6], where spatial encoding improved prediction accuracy compared to traditional models.

The objective of this study is to optimize the classification accuracy of large-scale tabular datasets by integrating effective feature selection methods with spatial transformation techniques, particularly through the TabMap algorithm. The motivation stems from the need to improve traditional ML models by introducing spatial encoding to help CNNs better capture intricate feature relationships, thereby improving prediction accuracy. Previous work on CNNs for tabular data, including the works of Matsuda et al. [5] and Zhu et al. [7], has shown that spatially transformed data enables CNNs to capture complex relationships that conventional ML models fail to recognize, especially in high-dimensional datasets.

This research is significant because it proposes a novel hybrid methodology that integrates feature selection with spatial transformation, enabling deep learning models to capture complex relationships in tabular data more effectively. The proposed methodology could have far-reaching implications in various domains, such as environmental monitoring and healthcare, where accurate classification of large-scale datasets is crucial. By demonstrating the effectiveness of TabMap and feature selection techniques, this study contributes to the growing field of deep learning for tabular data classification, showing the potential of CNNs beyond image and time-series data. Similar spatial encoding approaches, like Tab2vox [10] and HACNet [5], have also explored transforming tabular data into spatial representations, further validating the approach proposed in this paper.

The methodology consists of the following steps: first, a large dataset of freshwater quality is preprocessed to handle missing values, normalize features, and remove outliers. Then, feature selection methodologies such as Information Gain-based dynamic thresholding, Correlation-Based Feature Selection (CBFS), and others are utilized to select the best features. The preprocessed and selected features are converted into two-dimensional spatial grids using the TabMap algorithm. Convolutional Neural Network (CNN) models are trained on such transformed datasets, and the performance of such models is compared against that of conventional machine learning models like Random Forest, LightGBM, and XGBoost. Finally, the paper isstructured as follows: Section 1 introduces the topic of the paper and gives an overview of the research problem and the scope of the study. Section 2 gives a survey of the Related Works, engaging with the literature on feature selection and spatial transfor-

mation methodologies applied to tabular data. Section 3 outlines the Methods used including the dataset, preprocessing steps, feature selection methodologies, and the architecture of the CNN utilized in this work. Section 4 gives the Results or Findings of the paper, providing a comparison of the performance of the proposed approach with conventional models. Section 5 provides for the Discussion and includes the implications and limitations of this study. Finally, Section 6 concludes with the Conclusion and Future Work and summarizes key results and suggests avenues for future research.

1.2 RELATED WORKS

Recent advancements in deep learning have significantly influenced the analysis of tabular data, especially for large and high-dimensional datasets. The challenge lies in transforming traditional tabular data into formats that deep learning models, particularly Convolutional Neural Networks (CNNs), can process effectively. Various approaches have been proposed to address this challenge, with a growing emphasis on spatially encoding tabular data for use in CNNs. This literature review focuses on key contributions to tabular data analysis, including spatial transformation techniques, feature selection methods, and the application of CNNs to tabular data.

One of the strongest advances within this category is TabNet [1], a sequential attention-based neural model that learns how to select informative features for each of the on-the-go decision-making steps. TabNet achieved competitive performance across a variety of public tabular test sets for regression and classification tasks and frequently is better than gradient boosting-based solutions such as XGBoost. A notable redeeming quality of TabNet is its capability for global and local interpretability by means of feature attribution. TabNet is limited by its computationally expensive nature, which makes it frequently impractical for very large test sets. Beyond that, careful hyperparameter tuning is needed for the approach and is often tedious and resource-heavy. The limitations of TabNet create a rationale for more efficient methodologies such as the TabMap algorithm, which tries to couple feature choice with spatial transformations.

Contrary to TabNet, HACNet [5] integrates a hard-attention-based table-to-image converter and a CNN. The Gumbel-Softmax system is used by HACNet to transform features into pixels and consequently supports interpretability by allowing a feature to pixel one-to-one correspondence. The architecture of HACNet presented the strongest classification performance on various benchmark sets against other methods such as DeepInsight and IGTD. The advantage of HACNet is its capacity for simultaneously optimizing table-to-image conversion and CNN training. The challenge of complex joint optimization and the necessity for fine-tuning compensating loss terms contributes to scalability challenges for large-scale sets.

Another spatial encoding technique is DeepInsight [6], which uses dimensionality reduction methods like t-SNE and UMAP to encode tabular data into images. Two-dimensional embeddings of tabular data are created by DeepInsight by indexing features in such a way that the interrelationship of the features is preserved. The technique has shown to perform well on small to moderate-sized datasets, mostly

on fields involving genomics and healthcare application. Keep in mind that DeepInsight is highly computationally expensive, and its performance generally deteriorates with an increase in the dataset size. Further, by prioritizing feature relationships for preservation, but not considering the feature's individual importance, suboptimal embeddings sometimes occur in certain cases.

IGTD [7] offers another method for converting tabular data into spatial images, focusing on minimizing the distortion between feature correlations and pixel distances through iterative feature swaps. This method has been applied to gene expression data and molecular descriptors for anticancer drug response prediction. IGTD has proven effective in preserving feature similarities but is also computationally intensive, particularly for datasets with many features. The iterative nature of IGTD requires significant computational resources, which can be a drawback for large-scale applications.

The integration of feature selection with spatial encoding has been explored in several studies. Dynamic Weighted Tabular Mapping (DWTM) [8] computes feature weights based on their statistical correlations with class labels, allowing the algorithm to allocate more image space to important features. DWTM outperforms traditional machine learning models and other image embedding techniques such as SuperTML and IGTD. However, its performance is heavily reliant on accurate feature-target correlation estimates, which may not always be available or accurate, especially in noisy or sparse datasets.

Feature selection is an essential step in many machine learning pipelines, as it helps to reduce dimensionality and improve model interpretability. Information Gain (IG) is one of the most commonly used methods for feature selection, as it quantifies the relevance of a feature by measuring the amount of information it provides about the target variable. Prasetiyowati et al. [13] proposed a dynamic thresholding method for IG, which adapts based on the standard deviation of IG values, offering improvements in speed and accuracy for Random Forest classifiers on high-dimensional datasets. This adaptive thresholding approach has been proven effective in certain contexts but may not always generalize well, particularly when features exhibit weak or inconsistent relationships with the target variable.

One of the most significant contributions to the processing of tabular data is TabMap [9], a spatially semantic topographic map-based technique for transforming high-dimensional tabular data into 2D images. TabMap makes use of an optimal transportation algorithm in order to optimize the difference between feature correlation distance and the Euclidean distance of grid cells. The approach enables convolutional neural networks (CNNs) to learn of intricate patterns among features and thereby perform better in classification. TabMap has been shown to perform better than conventional approaches like XGBoost and Random Forest and more recent deep learning-based approaches like DeepInsight and IGTD. TabMap, however, mostly focuses on continuous data and suffers limitations in dealing with categorical characteristics or heterogeneous-type sets. The situation highlights the need for more resilient hybrid models that can handle a more varied set of data types, something that is tackled in this paper.

Although models like TabNet, HACNet, and TabMap exhibit powerful perfor-

mance and interpretability, limitations of computational complexity and scalability remain. Combining feature selection with spatial transformation, as attempted in this paper, tries to alleviate these limitations by reducing the dimension of the data before spatial encoding. By focusing the learning capability of the convolutional neural network (CNN) on the best-informative features, we try to increase both accuracy and efficiency, particularly for large-scale problems.

In short, the current literature describes that feature selection integrated with spatial encoding techniques improves tabular data classification. Nevertheless, the problems of computationally expensive cost, scalability, and how to treat categorical data persist. The paper offers to solve these issues by simultaneously combining TabMap and feature selection methods, and hence providing an efficient and interpretable solution for large-scale tabular data classification.

1.3 METHODS

This section describes the applied methodology of this work that integrates feature selection and spatial transformation for more effective classification on enormous table-based datasets. The methodology consists of data preprocesing, feature selection, spatial transformation by TabMap, and design of a Convolutional Neural Network (CNN) for facilitating classification. The overall methodology has been customized for enhancing freshwater quality classification by using a combination of approaches.

1.3.1 Research Methodology

The goal of this study is to improve the performance of classification tasks on large-scale tabular data by integrating feature selection and spatial transformations, specifically TabMap, along with deep learning practices. The solution entails various steps, such as the preprocessing of the dataset, using feature selection for reducing the dimensions, converting the data into spatial representations and, lastly, training a set of convolutional neural network models on the transformed sets. The performance of the CNN models is compared to base case machine learning models such as Random Forest, XGBoost, and LightGBM in a bid to ascertain the impact of the used practices of spatial transformation and feature selection. The adopted methodology's overall workflow is depicted in Fig. 1.1 and consists of the following major steps:

1.3.2 Dataset and Data Preprocessing

The dataset used in this study is a large-scale freshwater quality dataset containing 4.5 million instances and 23 features. These features include water quality indicators such as pH, iron concentration, air temperature, and nitrate levels. The data undergoes several preprocessing steps to ensure its consistency and quality. Missing values are imputed using the median for continuous variables and the mode for categorical variables. Normalization is performed on continuous features using Min-Max scaling, and categorical variables are one-hot encoded. Outliers are identified using the Z-score method and removed to maintain the integrity of the dataset. These preprocessing

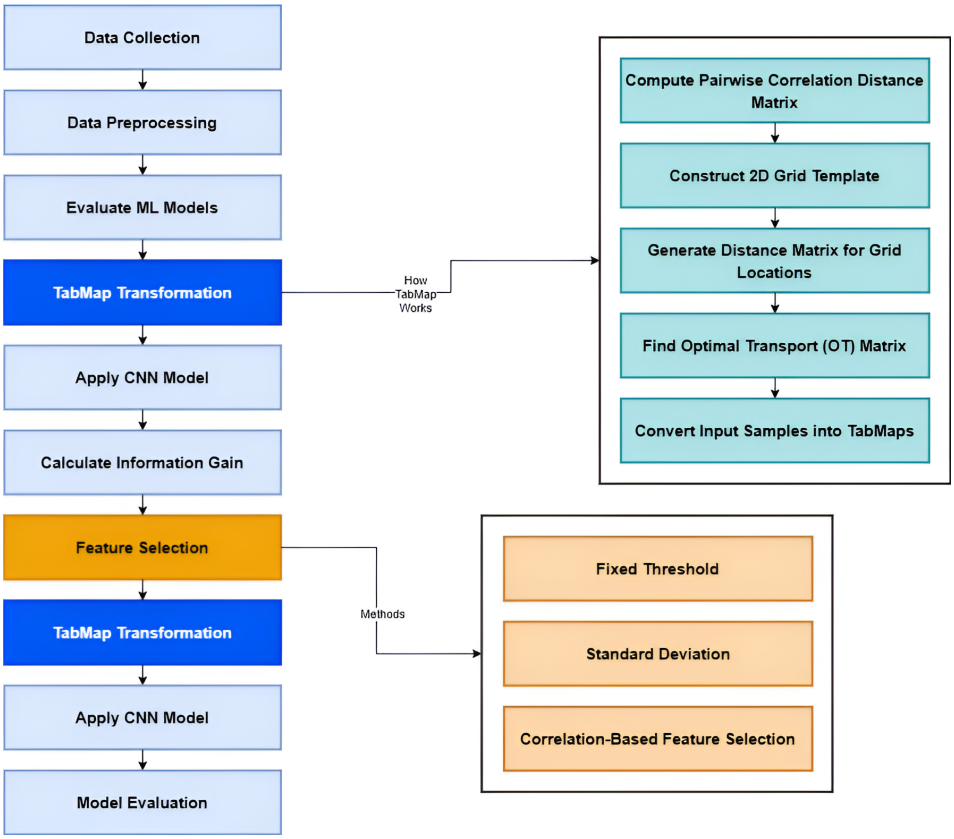


Figure 1.1: Overall methodology workflow.

steps ensure that the dataset is clean and standardized, making it suitable for the subsequent analysis.

1.3.3 Baseline Machine Learning Model Training

Seven established machine learning models were trained and optimized using grid search and cross-validation to establish baseline performance on the raw tabular data. These models included Random Forest (RF), Light Gradient Boosting Machine (LightGBM), Categorical Boosting (CatBoost), Adaptive Boosting (AdaBoost), Extreme Gradient Boosting (XGBoost), Gradient Boosting Classifier (GBC), and Decision Tree (DT). Each model was trained using default settings initially, and hyperparameters were further tuned using grid search to identify the best configuration. The models' performance was evaluated based on accuracy and F1-score, providing a robust comparison across various machine learning techniques.

1.3.4 TabMap Spatial Transformation

The key innovation in this study is the use of TabMap for transforming high-dimensional tabular data into a spatially meaningful 2D grid. This transformation

preserves the feature correlations and organizes them spatially, allowing the CNN to exploit the spatial locality of the data. The TabMap algorithm computes pairwise feature correlation distances and uses an optimal transport framework to map these features onto a 2D grid. Features that are highly correlated are placed closer together, preserving the inherent relationships in the data. This transformation allows the CNN to recognize spatial dependencies and relationships between features, which traditional ML models might overlook.

1.3.5 TabMap Representation

Once the dataset is transformed into a spatial representation using TabMap, each feature is represented as a grid cell. The value of each cell is determined by the feature's value in the original tabular data. This 2D representation serves as input to the CNN. The transformed data is a visually interpretable grid, where the intensity of each grid cell corresponds to the magnitude of the feature. This representation enables CNNs to process tabular data as if it were an image, using their well-established ability to capture spatial features and patterns.

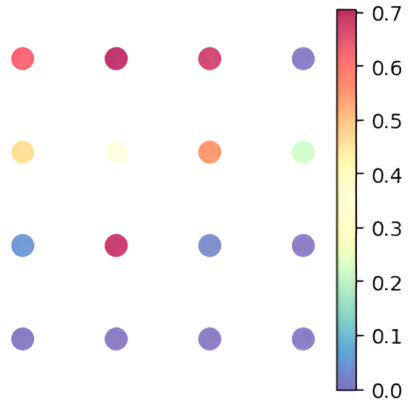


Figure 1.2: Example of a TabMap 2D Grid.

1.3.6 Feature Selection

Feature selection is an important step in the dimension reduction process of the dataset and simultaneously enhancing the interpretability of the model. The present study employs three different feature selection methodologies: Information Gain, Correlation-Based Feature Selection (CBFS), and an adaptive Information Gain-based dynamic thresholding method. The Information Gain methodology rank orders the features by the amount of information they contribute regarding the target variable. The CBFS methodology eliminates highly correlated features and, as a result, retains the most informative and independent features. The adaptive Information Gain-based dynamic thresholding methodology adapts the threshold for feature selection based on the standard deviation of the Information Gain measures and, as such, enables improvements in classifier speed and efficiency. The thusly selected fea-

tures facilitate the production of the TabMap spatial representations and, as such, ensure that the Convolutional Neural Network (CNN) is trained on the best relevant and informative data.

1.3.7 CNN Architecture

The Convolutional Neural Network (CNN) architecture used in the current examination is specifically designed for processing the two-dimensional grid forms of table-based data. The network consists of several convolutional layers followed by pooling layers allowing for feature extraction from the spatial representations. The convolutional layers use filters on the input grids and hence extract low-level features like patterns and edges. Max-pooling layers are used subsequent to every convolutional layer for downsampling the feature maps and reducing their spatial dimensions considerably with the retention of significant information. The technique helps in avoiding overfitting and reducing the model's computational intensity.

The feature maps from the pooling layers are then flattened and passed through fully connected layers, which capture higher-level patterns and make predictions. The output layer uses a softmax activation function for binary classification, outputting the probability of each class. Rectified Linear Unit (ReLU) is used as the activation function in both convolutional and fully connected layers to introduce non-linearity into the model, allowing it to learn more complex relationships. Dropout regularization is applied to the fully connected layers to prevent overfitting by randomly deactivating certain neurons during training.

The CNN is trained using the Adam optimizer, which adapts the learning rate during training to improve convergence. Categorical cross-entropy is used as the loss function for the classification task. The model's performance is evaluated using accuracy and F1-score.

1.3.8 Model Evaluation

The performance of the proposed CNN model is evaluated using two performance metrics: accuracy and F1-score. These metrics are calculated for both the overall model and for each individual class. The CNN model is compared against the baseline machine learning models, such as Random Forest, XGBoost, and LightGBM, to assess the improvements achieved by spatial transformation and feature selection. Additionally, the interpretability of the CNN model is assessed by analyzing the learned filters and feature maps, which provide insights into which features the model considers most important for making predictions.

1.4 RESULTS OR FINDINGS

This section analyzes the results of the proposed hybrid framework combining TabMap, feature selection, and CNNs. It highlights performance improvements over traditional models and the impact of feature selection on classification accuracy. The experimental findings demonstrate that spatial encoding of tabular data enables convolutional neural networks to capture complex feature interactions more effectively

than conventional machine learning algorithms. Moreover, applying feature selection prior to spatial transformation not only reduces computational complexity but also enhances model focus on relevant attributes, resulting in significant gains in predictive performance. These results corroborate and extend previous research emphasizing the importance of combining feature selection with spatial data representations for improved tabular data classification.

1.4.1 Performance Comparison: Traditional Models versus CNN on TabMap Data

Table 1.1: Performance metrics comparison between traditional machine learning models and CNN trained on TabMap-transformed data (before feature selection).

Model	**Accuracy**	**F1 Score**
LightGBM	0.8717	0.8758
CatBoost	0.8649	0.8686
Gradient Boosting	0.8348	0.8348
AdaBoost	0.7980	0.7867
XGBoost	0.8616	0.8651
Random Forest	0.8789	0.8827
Decision Tree	0.8191	0.8187
CNN (Before Feature Selection)[After TabMap]	**0.9008**	**0.9061**

The results clearly demonstrate in Table 1.1 that the CNN model trained on TabMap images significantly outperforms all traditional models, achieving an accuracy of 90.08% and an F1-score of 90.61%. This highlights the advantage of spatially encoding tabular features in a way that leverages the CNN's capability to capture spatial dependencies, improving classification performance on complex, large-scale data.

1.4.2 Impact of Feature Selection on CNN Performance

Applying feature selection further enhances model accuracy and efficiency. Table 1.2 details the performance of CNN models trained on TabMap images generated from feature-selected subsets using the three selection methods.

Table 1.2: CNN model performance after applying feature selection methods and TabMap transformation.

Model	**Accuracy**	**F1 Score**
CNN (Fixed Threshold Feature Selection)	0.9312	0.9300
CNN (Standard Deviation Feature Selection)	0.9276	0.9257
CNN (Correlation-Based Feature Selection)	**0.9317**	**0.9304**

Among the three, the Correlation-Based Feature Selection (CBFS) method yields the best results, improving accuracy to 93.17% and F1-score to 93.04%. These improvements underscore the importance of reducing redundant and irrelevant features

before spatial transformation, allowing the CNN to focus on the most discriminative aspects of the data.

In summary, the results validate that the TabMap spatial transformation enables CNNs to extract meaningful feature interactions in tabular data, resulting in superior classification performance over traditional machine learning models. Moreover, integrating feature selection methods prior to spatial transformation significantly improves accuracy and efficiency, with CBFS providing the highest gains. These findings confirm the effectiveness of the proposed methodology for large-scale freshwater quality classification and offer a promising direction for similar tabular data challenges in environmental science and beyond.

1.5 DISCUSSION

This work verifies the primary hypothesis that combining feature selection with spatial transformation using TabMap greatly improves the classification accuracy of Convolutional Neural Networks (CNNs) applied to tabular data. By transforming tabular data into spatial representations, TabMap enables CNNs to recognize subtle relationships among features difficult to perceive using traditional machine learning models. The accuracy of 90.08% and F1-score of 90.61% were achieved by CNNs trained on TabMap-transformed images, significantly better than tree-based models like Random Forest, LightGBM, and CatBoost. After feature selection was introduced, the accuracy improved to 93.17%, and Correlation-Based Feature Selection (CBFS) provided the best performance. This highlights the effectiveness of removing redundant features in boosting the capability of the model for learning from high-dimensional data. The current results are in accord with the results shown by Yan et al. [9], and thereby verifies the effectiveness of TabMap on a series of different fields of application, as well as those by Prasetiyowati et al. [13], which pointed to the value of dynamic feature filtering in enhancing model performance.

Our results also align with prior work by Lee et al. [2] and Sharma et al. [6], which demonstrated that spatial encoding techniques benefit CNNs by capturing spatial dependencies and relationships between features. A novel contribution of this study is the systematic comparison of multiple feature selection strategies within the TabMap-CNN framework. The evaluation reveals that the combination of spatial transformation and feature selection significantly impacts the model's ability to improve classification accuracy.

An unexpected finding in this study was that adaptive thresholding using standard deviation performed worse than the fixed and correlation-based methods for feature selection. This suggests that removing correlation-based redundancy may be more effective for datasets with complex relationships between features, such as those found in environmental data. Additionally, the strong performance of CNNs even before feature selection was applied indicates that TabMap alone can significantly enhance the learning process by preserving the spatial relationships among features, as seen in the improvements observed even before any feature selection was performed.

Despite these promising results, several limitations and challenges must be addressed. One limitation is the use of one-hot encoding for categorical variables, which

may limit the effectiveness of spatial encoding, as it increases the dimensionality of the data and could introduce sparsity issues. Future work should explore alternative encoding strategies that better preserve the semantic meaning of categorical variables, such as embedding techniques, which have been shown to handle categorical data more efficiently. Additionally, while the results were promising for the freshwater quality dataset used in this study, broader validation across multiple domains is necessary to determine the generalizability and robustness of the proposed framework. Testing the approach on datasets from different domains, such as healthcare or finance, would help assess the applicability of the method in various real-world contexts.

These computational demands of TabMap and Convolutional Neural Networks (CNNs) do present serious issues of scalability. The more instances and features in the dataset, the more likely the computational expense of TabMap transforming and training of CNNs will become impracticable. This is particularly so for large sets of data with complex feature relationships. Additional effort needs to focus on optimising the TabMap transforming process by considering parallelising or approximate methodologies, as these would enhance efficiency and render the approach more scalable for large sets of data.

Finally, although the present study is interested in classification issues, the framework described can potentially be generalized for regression issues, in which continuous target measures have to be predicted. An examination of TabMap and feature selection performance for regression issues could potentially broaden the scope of the approach still further and yield useful results for a wider range of data-based applications. More sophisticated neural network architectures, including attention-based models or the Graph Neural Networks (GNNs), may similarly enhance performance by identifying more sophisticated relationships in the data.

In conclusion, this study demonstrates the success of blending spatial encoding and feature selection for better large-scale tabular dataset classification. The system not only ensures better accuracy and interpretability but also offers a good foundation for future developments on TabMap's scalability boost, efficient processing of categorical information, regression task support, and neural network models of greater sophistication.

1.6 CONCLUSION AND FUTURE WORK

This paper presented a hybrid model by integrating feature selection and spatial transformation with TabMap algorithm and Convolutional Neural Networks (CNNs) for boosting the classification of large tabular data. The driving force was overcoming traditional models' inefficiencies in effectively capturing intricate feature relationships in high-dimensional data.By converting tabular data into spatially coherent 2D images that preserve feature correlations, TabMap enables CNNs to learn richer representations, improving model performance.

Experiments on a large freshwater quality dataset (4.5 million instances, 23 features) demonstrated that CNNs trained on TabMap images outperformed baseline models such as Random Forest, LightGBM, and CatBoost in both accuracy and F1-

score. TabMap's spatial encoding effectively captured both local and global feature interactions that traditional models often miss. Feature selection using Information Gain further improved the model's performance, with Correlation-Based Feature Selection (CBFS) yielding the best results by reducing feature redundancy before spatial mapping.

While the proposed approach showed promising results, it has limitations. TabMap is primarily suited for continuous features, and its performance on datasets with a large number of categorical features may be constrained. The evaluation was domain-specific, and broader validation is needed across different datasets and use cases. Additionally, the CNN architecture used was standard; future work could explore more advanced models, such as attention-based networks or graph neural networks (GNNs), to further improve performance. Computational cost remains a concern, particularly for large-scale deployment, and optimizing the TabMap transformation process would enhance scalability.

Looking forward, future directions include improving the handling of categorical and ordinal data, extending the framework to regression tasks, and integrating more advanced neural architectures. The methodology could also be applied to other domains, such as healthcare, finance, and retail, to assess its generalizability and effectiveness in real-world applications. This research lays a strong foundation for applying deep learning to tabular data, offering a scalable and interpretable approach to classification. Additionally, the Mamba architecture, as described in greater detail in the thorough survey by Salam et al., provides a potential direction along which model performance on a range of applications ranging from natural language processing (NLP), computer vision, and time-series analysis can be improved. Its selective state-space modeling and incorporation of cutting-edge methods in the form of transformers hold potential in forestalling the computationally limiting bottlenecks often characteristic of large-scale tabular data classification. Its efficient memory handling and lower computation, compared to classical models, make it a candidate worth exploring in future research. Future work could explore how Mamba could be applied on large, wide datasets, with a specific focus on increasing scalability and efficient categorical features handling [4].

1.7 ACKNOWLEDGEMENT

This work has been performed in AIUB. The authors thank AIUB authority for their financial and any other support.

Bibliography

[1] Arik, S. O., and Pfister, T.: TabNet: Attentive Interpretable Tabular Learning. arXiv:1908.07442 [cs, stat]. Available: https://arxiv.org/abs/1908.07442 [Accessed: Mar. 3, 2025]

[2] Lee, E., Nam, M., and Lee, H.: Tab2vox: CNN-Based Multivariate Multilevel Demand Forecasting Framework by Tabular-To-Voxel Image Conversion. Sustainability, 14(18), 11745. https://doi.org/10.3390/su141811745.

[3] Gómez-Martínez, V., Lara-Abelenda, F. J., Peiro-Corbacho, P., Chushig-Muzo, D., and Soguero-Ruiz, C.: LM-IGTD: a 2D image generator for low-dimensional and mixed-type tabular data to leverage the potential of convolutional neural networks. Available: https://www.researchgate.net/publication/381652764_LM-IGTD_a_2D_image_generator_for_low-dimensional_and_mixed-type_tabular_data_to_leverage_the_potential_of_convolutional_neural_networks. [Accessed: Mar. 9, 2025]

[4] Salam, A., Mahmud, R., Islam, T., Mukta, S., and Shatabda, S.: A Comprehensive Survey on Mamba: Architectures, Challenges, and Opportunities. Available: https://doi.org/10.1109/MC.2025.3571322.

[5] Matsuda, T., Uchida, K., Saito, S., and Shirakawa, S.: HACNet: End-to-end learning of table-to-image converter and convolutional neural network. https://doi.org/10.21203/rs.3.rs-2174672/v1.

[6] Sharma, A., López, Y., Jia, S., Lysenko, A., and Tsunoda, T.: Enhanced analysis of tabular data through Multi-representation DeepInsight. Scientific Reports, 14(1). https://doi.org/10.1038/s41598-024-63630-7.

[7] Zhu, Y., et al.: Converting tabular data into images for deep learning with convolutional neural networks. Scientific Reports, 11(1). https://doi.org/10.1038/s41598-021-90923-y.

[8] Iqbal, M. I., Hossain, S., Hasan, A. R., and Islam, S.: A Dynamic Weighted Tabular Method for Convolutional Neural Networks. IEEE Access, vol. 10, pp. 134183–134198, Jan. 2022. https://doi.org/10.1109/access.2022.3231102.

[9] Yan, R., Islam, M. T., and Xing, L.: Interpretable discovery of patterns in tabular data via spatially semantic topographic maps. Nature Biomedical Engineering. https://doi.org/10.1038/s41551-024-01268-6.

[10] Lee, S., and Oh, S.: Table2Image: Interpretable Tabular Data Classification with Realistic Image Transformations. Available: https://www.researchgate.net/publication/386577285_Table2Image_Interpretable_Tabular_data_Classification_with_Realistic_Image_Transformations. [Accessed: Mar. 23, 2025]

[11] Medeiros Neto, L., Neto, S., and Endo, P. T.: A comparative analysis of converters of tabular data into image for the classification of Arboviruses using Convolutional Neural Networks. PLoS ONE, 18(12), e0295598. https://doi.org/10.1371/journal.pone.0295598.

[12] Wolf, T. N., Pölsterl, S., and Wachinger, C.: DAFT: A universal module to interweave tabular data and 3D images in CNNs. NeuroImage, 260, 119505. https://doi.org/10.1016/j.neuroimage.2022.119505.

[13] Prasetiyowati, M. I., Maulidevi, N. U., and Surendro, K.: Determining threshold value on information gain feature selection to increase speed and prediction

accuracy of random forest. Journal of Big Data, 8(1). https://doi.org/10.1186/s40537-021-00472-4.

[14] Zhao, M., Cao, G., Huang, X., and Yang, L.: Hybrid Transformer-CNN for Real Image Denoising. IEEE Signal Processing Letters, vol. 29, pp. 1252–1256, Jan. 2022. https://doi.org/10.1109/lsp.2022.3176486.

[15] Alabsi, B. A., Anbar, M., and Rihan, S. D. A.: CNN-CNN: Dual Convolutional Neural Network Approach for Feature Selection and Attack Detection on Internet of Things Networks. Sensors, vol. 23, no. 14, p. 6507, Jan. 2023. https://doi.org/10.3390/s23146507.

[16] Zhang, Q., Xiao, J., Tian, C., Chun-Wei Lin, J., and Zhang, S.: A robust deformed convolutional neural network (CNN) for image denoising. CAAI Transactions on Intelligence Technology, Jun. 2022. https://doi.org/10.1049/cit2.12110.

[17] Zhou, W., Wang, H., and Wan, Z.: Ore Image Classification Based on Improved CNN. Computers and Electrical Engineering, vol. 99, p. 107819, Apr. 2022. https://doi.org/10.1016/j.compeleceng.2022.107819.

[18] Damri, A., Last, M., and Cohen, N.: Towards efficient image-based representation of tabular data. Neural Computing and Applications, vol. 36, no. 2, pp. 1023–1043, Oct. 2023. https://doi.org/10.1007/s00521-023-09074-y.

[19] Borisov, V., Leemann, T., Sessler, K., Haug, J., Pawelczyk, M., and Kasneci, G.: Deep Neural Networks and Tabular Data: A Survey. IEEE Transactions on Neural Networks and Learning Systems, pp. 1–21, 2022. https://doi.org/10.1109/tnnls.2022.3229161.

[20] Yang, J., et al.: MedMNIST v2 - A large-scale lightweight benchmark for 2D and 3D biomedical image classification. Scientific Data, vol. 10, no. 1, p. 41, Jan. 2023. https://doi.org/10.1038/s41597-022-01721-8.

[21] Dhal, P., and Azad, C.: A comprehensive survey on feature selection in the various fields of machine learning. Applied Intelligence, vol. 52, Jul. 2021. https://doi.org/10.1007/s10489-021-02550-9.

[22] Jain, V., Goel, M., and Shah, K.: Deep Learning on Small Tabular Dataset: Using Transfer Learning and Image Classification. Communications in Computer and Information Science, pp. 555–568, 2022. https://doi.org/10.1007/978-3-030-95711-7_46.

[23] Erickson, N., Shi, X., Sharpnack, J., and Smola, A.: Multimodal AutoML for Image, Text and Tabular Data. Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Aug. 2022. https://doi.org/10.1145/3534678.3542616.